CSE 111 Fall 2026: Project Checkpoint 1

F1 Fan Analytics Database

Eliud Hadgu, Nathan Delgado

Synopsis

This project is a relational database that brings F1 data into one structured, queryable system, modeling drivers, teams, circuits, races, seasons, sponsors, and engine suppliers along with the relationships between them, such as driver transfers, race-by-race results, and sponsorship and engine supply deals. Because these changes are tracked over time, fans search which drivers perform best at a given circuit, how two teammates compare across a season, or how a team's results changed after switching engine suppliers. The design is also built to grow, with future additions like lap times, pit stops, and tire data, and a web dashboard that could give fans an interactive way to explore the data.

Authors & Contribution: Eliud 100%

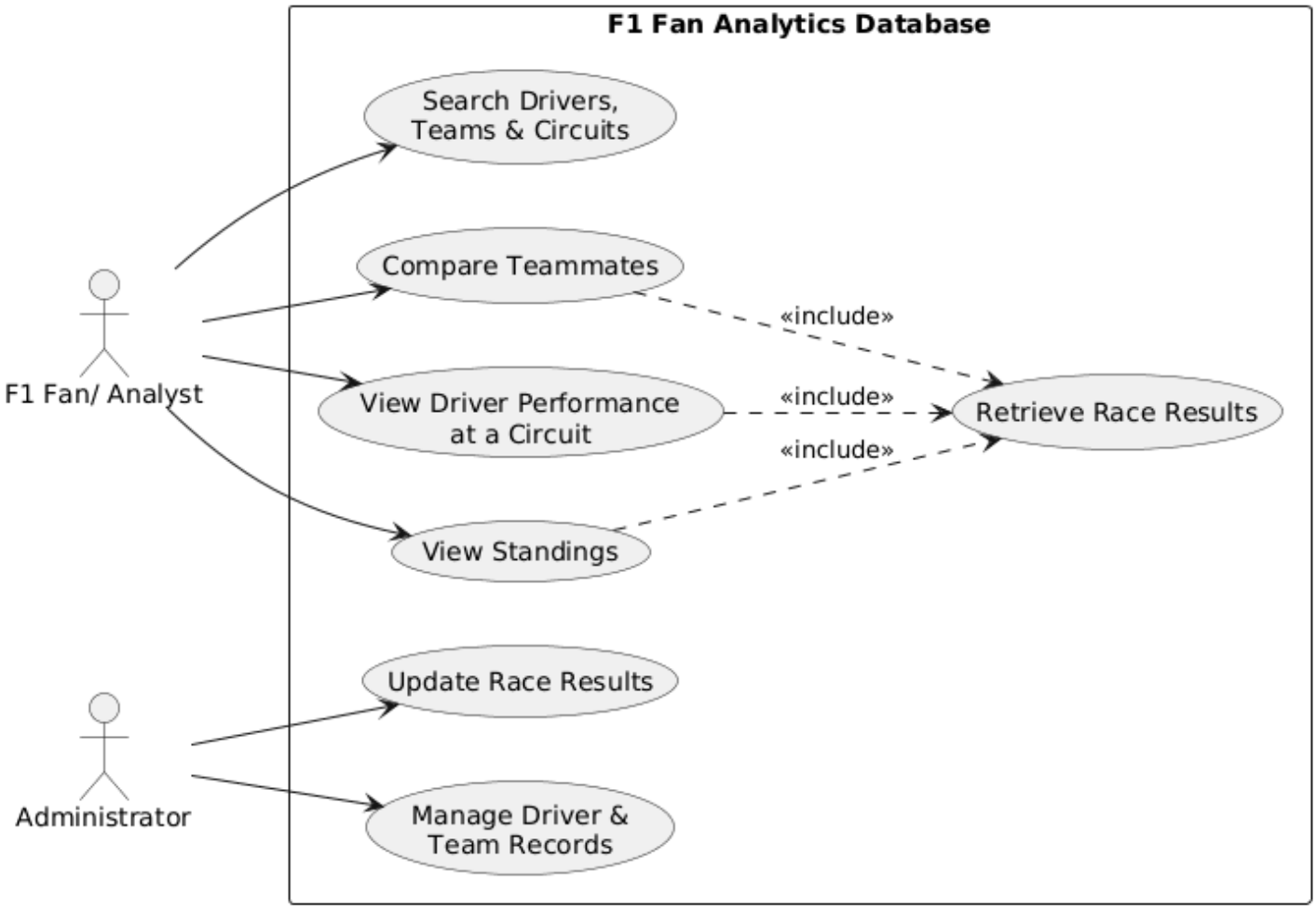
UML Use Case Diagram

The primary use case modeled in this system is allowing F1 fans and analysts to search and analyze Formula 1 racing data. Users can search for drivers, teams, and circuits, compare teammates across seasons, view driver performance at specific circuits, and access championship standings. These functions retrieve race results from the database to display relevant statistics, including finishing positions, points earned, and team performance.

The system also includes an Administrator responsible for maintaining accurate and updated information. Administrators can update race results and manage driver and team records, ensuring the database remains reliable. Overall, the system provides a structured way to explore historical F1 data, compare performance, and track racing statistics across multiple seasons.

Authors & Contribution: Nathan 100%

Use Case Diagram:

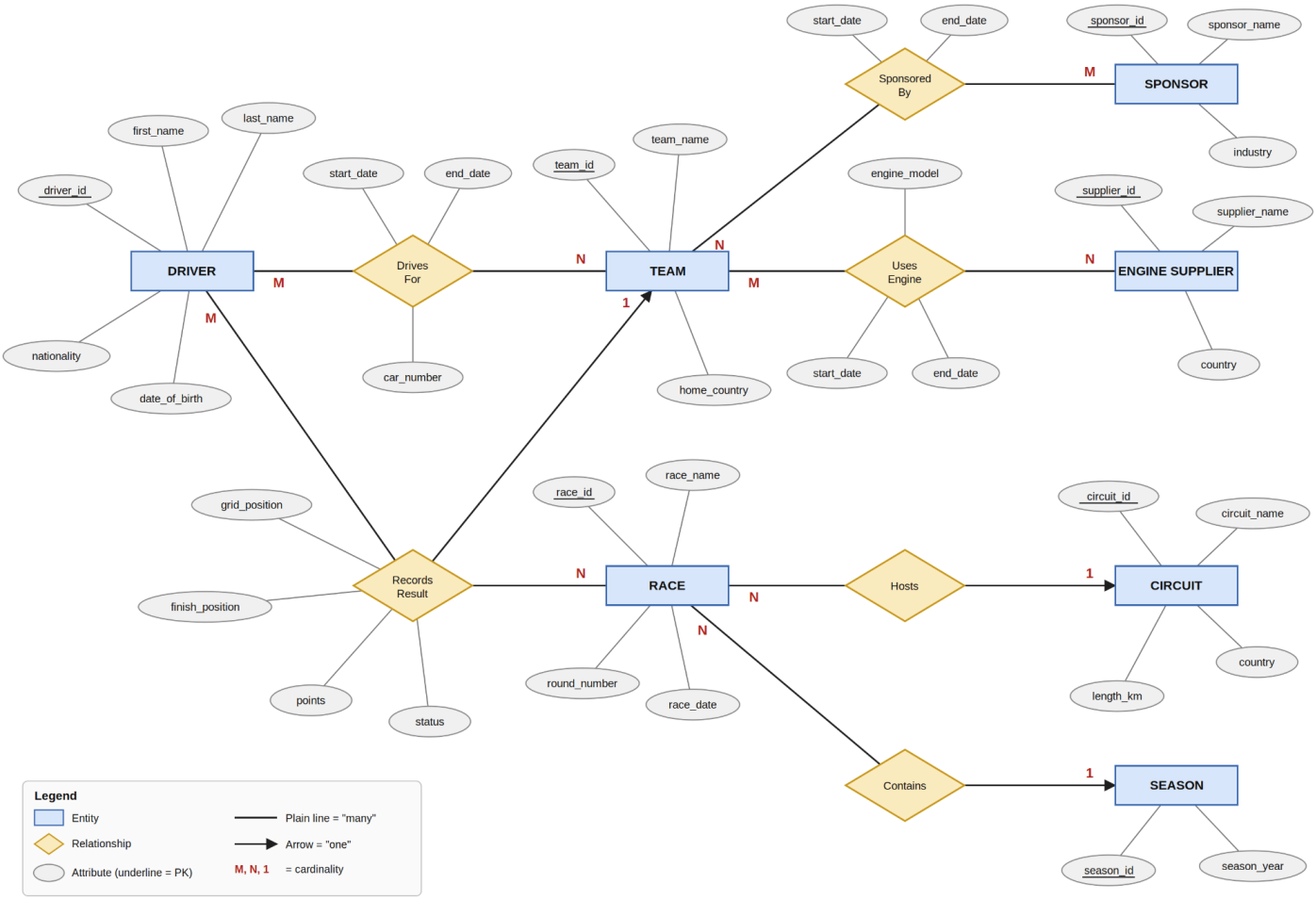


Authors & Contribution: Eliud 100%

ER Diagram

The database consists of 7 entities (Driver, Team, Circuit, Race, Season, Sponsor, and Engine Supplier) connected through 6 relationships: Drives For (many-to-many between Drivers and Teams), Sponsored By (many-to-many between Teams and Sponsor), Uses Engine (many-to-many between Teams and Engine Suppliers), Records Result (between Drivers, Teams, and Races), Hosts (one-to-many between Circuits and Races), and Contains (one-to-many between Seasons and Races).

ERD-1:



Authors & Contribution: Nathan 100%

(Add more ER diagrams as needed and number them sequentially with contribution listed under each)

Relational Schema

Driver

```
CREATE TABLE Driver (
    driver_id INTEGER PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    nationality VARCHAR(50),
    date_of_birth DATE
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Team

```
CREATE TABLE Team (
    team_id INTEGER PRIMARY KEY,
    team_name VARCHAR(100) NOT NULL,
    home_country VARCHAR(50)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Race

```
CREATE TABLE Race (
    race_id INTEGER PRIMARY KEY,
    race_name VARCHAR(100) NOT NULL,
    round_number INTEGER,
    race_date DATE,
    circuit_id INTEGER NOT NULL,
    season_id INTEGER NOT NULL,
    FOREIGN KEY (circuit_id) REFERENCES Circuit(circuit_id),
    FOREIGN KEY (season_id) REFERENCES Season(season_id)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Sponsor

```
CREATE TABLE Sponsor (
    sponsor_id INTEGER PRIMARY KEY,
    sponsor_name VARCHAR(100) NOT NULL,
    industry VARCHAR(50)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Engine Supplier

```
CREATE TABLE Engine-Supplier (
    supplier_id INTEGER PRIMARY KEY,
    supplier_name VARCHAR(100) NOT NULL,
    country VARCHAR(50),
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Circuit

```
CREATE TABLE Circuit (
    circuit_id INTEGER PRIMARY KEY,
    circuit_name VARCHAR(100) NOT NULL,
    country VARCHAR(50),
    length_km DECIMAL(5,3)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Season

```
CREATE TABLE Season (
    season_id INTEGER PRIMARY KEY,
    season_year INTEGER NOT NULL UNIQUE
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Drives For

```
CREATE TABLE Drives_For (
    driver_id INTEGER,
    team_id INTEGER,
    start_date DATE,
    end_date DATE,
    car_number INTEGER,
    PRIMARY KEY (driver_id, team_id, start_date),
    FOREIGN KEY (driver_id) REFERENCES Driver(driver_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Sponsored By

```
CREATE TABLE Sponsored_By (
    sponsor_id INTEGER,
    team_id INTEGER,
    start_date DATE,
    end_date DATE,
    PRIMARY KEY (sponsor_id, team_id, start_date),
    FOREIGN KEY (sponsor_id) REFERENCES Sponsor(sponsor_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%

Uses Engine

```
CREATE TABLE Uses_Engine (
    team_id INTEGER,
    supplier_id INTEGER,
    start_date DATE,
    end_date DATE,
    engine_model VARCHAR(50),
    PRIMARY KEY (team_id, supplier_id, start_date),
    FOREIGN KEY (team_id) REFERENCES Team(team_id),
    FOREIGN KEY (supplier_id) REFERENCES Engine_Supplier(supplier_id)
);
```

Authors & Contribution: Eliud 50%, Nathan 50%